UPC Station Review 2019-2025

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Objectives

- Evaluate hourly and daily data for Upper Penticton Creek weather stations for 2019 to late winter/spring 2025
- Indicate missing or bad data and revise these data where feasible.
- Provide original and adjusted data in spreadsheets.

Methods

Hourly and daily data for each station were added to spreadsheets based on the calendar year. Data were evaluated graphically using daily and hourly time series of variables. Erroneous or missing data were noted and are indicated by NAN. Some issues are related to errors induced during reprogramming the data loggers and cannot be corrected.

Relative humidity was adjusted as necessary so that the maximum value did not exceed 100%. Snow depth and daily precipitation as snow (mm) were calculated from the sonic ranging sensor data. Incident solar radiation was adjusted where possible in the winter for snow on the sensor. An algorithm was added to hourly files to evaluate shading and sensitivity drift of the solar sensor. The 4Seasons pressure transducers and the Geonor precipitation gauge signals were filtered to remove spikes and other noise where possible in calculation of hourly and daily precipitation. The temperature and humidity data were used to calculate vapour pressure and vapour pressure deficit. The 0.005m soil temperature, and the 0.2 m and 0.5 m snow temperature data were used to estimate an average value for snow temperature. An evaluation of each variable at stations P1, P5, PB, PC, PJ, PK, PL and Geonor is presented below.

Data archive: All adjustments are noted in the spreadsheets and can be rejected if necessary. There are separate daily and hourly spreadsheets each year for P1, P5, PB, PC, PJ and PK (STXXXXD.xlsx and STXXXXH.xlsx where ST is the station, XXXX is the year and D and H indicate daily and hourly data, respectively). PL and the Geonor each have one spreadsheet per year with daily and hourly data as worksheets (PLXXXXD_H.xlsx and GeonorXXXD_H.xlsx, where XXXX is the year). Measured and adjusted values are in the Data worksheet. A summary value of each variable suitable for further analysis is included in the Data worksheet. A Meta worksheet lists equipment changes, measurements and issues with the data for that year. Graphs of daily time series are in the Graph worksheet. There are half-hour time series graphs as worksheets for the 4Seasons and Geonor indicating raw and filtered data.

The summary values from the daily spreadsheet were combined for each station to provide a time series from 2019 to 2055. The file name is ST_2019-2025DSumry.xlxs where ST is the station name (P1, P5, PB, PC, PJ, PK, and PL_Geonor) with the data in the Data worksheet. The ability to review variables graphically in the Graph worksheet and the Meta worksheet briefly describes the file. The files are in the

folder Sumry. csv files for each station and a program in R to review and compare variables are in a folder within the Sumry folder.

Equipment:

- Data logger: Campbell Sci CR10X (PK to 2021), CR1000 (P1, P5, PB, PC, PJ, PK) and CR300 (Geonor), and Hobo micro station (PL)
- Air temperature: CSI HMP45C (P5, PJ, PK), HC-S3 (P1, PB, PC) or HMP155A (P5, PJ, PK, 2021 onwards), Hobo temperature sensor (PL).
- Relative humidity: CSI HMP45C (P5, PJ, PK), HC-S3 (P1, PB, PC) or HMP155A (P5, PJ, PK, 2021 onwards), Hobo humidity sensor (PL).
- Surface temperature: Apogee infra-red thermometer (P1, P5, PB, PC, PJ, PK)
- Snow temperature: Chromel-constantan thermocouples referenced to thermistor on panel (P1, P5, PB, PC, PJ, PK)
- Soil temperature: Chromel-constantan thermocouples referenced to thermistor on panel (P1, P5, PB, PC, PJ, PK)
- Snow depth: Campbell Sci. SR50A sonic ranger (P1, P5, PB, PC, PJ, PK), or Sommer ultrasonic distance sensor (PL)
- Rainfall: Sierra Misco (P1, P5, PB, PC, PJ, PK), Texas Electronic (P1 2021 onwards) and Hobo (PL) tipping buckets
- Precipitation: 4 Seasons standpipe gauge with Sensotech and Stevens hydroprobe pressure transducers (P1) and Geonor T200B precipitation gauge (Geonor)
- Solar radiation - incident: LiCor LI200 silicon pyranometer (P1, PB, PJ, PK) and Hobo silicon pyranometer (PL)
- Solar radiation - reflected: LiCor LI200 silicon pyranometer (PB, PK) with adjustment for spectral response to reflected radiation, Epply B&W pyranometer (PJ)
- Soil moisture: CS616 Time domain reflectometer (0-0.3 m depth?) (P1, P5, PB, PC, PJ, PK) and Hobo TDR at 0.1 and 0.2 m depths (PL).
- Snowmelt lysimeter: Homemade tipping buckets and plywood box (P1).
- Wind speed: MetOne anemometer (PK to 2024), RMYoung wind monitor (PB, PJ and PK 2024 onwards) and Hobo wind speed sensor (PL)
- Wind direction: RMYoung wind monitor (PB, PJ and PK 2024 onwards), and Hobo wind direction sensor (PL)

Evaluation of Station Records

P1

- Air temperature at ~3 m: Looks good.
- Relative humidity at ~3 m: Looks good. No adjustment made for drift in sensitivity.
- Apogee IRT surface temperature: Failed in 2018 with intermittent values. Replaced Oct 2023. Not behaving as it should when compared to air temperature.
- Snow temperature at 0.2 and 0.5 m heights: Look good. Average snow temperature calculated.

- Soil temperature at 0.005, 0.15 and 0.5 m depths: 0.005 and 0.15 m depths look OK. The 0.5 m sensor is not working (likely disconnected). Lost in 2021 when the soil moisture probe was added.
- Snow depth: Looks good. Preliminary estimate of daily and hourly depths need filtering for signal noise due to thermal effects, blowing snow, rain etc.
- Rain gauge: Replacement of gauge in June 2021 did not include program change to new sensitivity (was 0.254 mm/tip, should be 0.1 mm/tip). Correct sensitivity is applied to data in spreadsheet.
- 4 Seasons precipitation gauge: Stevens pressure transducer for depth of water failed in 2023. Its water temperature looks OK, as does the independent thermocouple. The 4Seasons Sensotech pressure transducer is OK (starts in June 2019). Failure to empty and recharge before overflow has resulted in missing data during some years. A preliminary estimate of hourly and daily precipitation was obtained by filtering the signal to resolve variation due to thermal and pressure effects on sensors.
- Solar radiation - incident: Missing from daily output starting in 2021. Hourly data are available and used to calculated daily values. Values are too low Feb 2022 to Nov 2023 (possibly wrong sensitivity?). There is a large negative offset continuing on to 2025. Reduced values in winter due to snow on sensor and there are no reflected data to aid correction for this. An algorithm was added to the hourly files to evaluate shading and sensitivity drift of sensor.
- Soil moisture: A TDR probe was added in June 2021 and I failed (or was disconnected) July 2024.
- Wind speed at 3 m: Looks good. May be decreasing over time as surrounding forest canopy increases in height.
- Snowmelt lysimeters with tipping buckets: Blockages affecting functionality. Daily values affected by poor drainage and do not matching daily precipitation in summer. Unreliable measurement.
- Vapour pressure and vapour pressure deficit: Calculated from air temperature and relative humidity on hourly and daily basis.

P5

- Air temperature at ~3 m: Intermittent failure started in 2021 after the change to the HMP155A. Incorrect average and minimum temperature deleted.
- Relative humidity at ~ 3m: Intermittent failure started in 2021 after the change to the HMP155A. Incorrect average and minimum relative humidity deleted. No adjustment made for a drift in sensitivity.
- Apogee IRT surface temperature: Looks good.
- Snow temperature at 0.2 and 0.5 m heights: Look good. Average snow temperature calculated.
- Soil temperature at 0.005, 0.15 and 0.5 m depths: Look good.
- Snow depth: Looks good. Preliminary estimate of daily and hourly depths need filtering for signal noise due to thermal effects, blowing snow, rain etc.
- Soil moisture: A TDR probe was added in June 2021. Data look good.

- Vapour pressure and vapour pressure deficit: Calculated from air temperature and relative humidity on hourly and daily basis.

PB

- Air temperature at ~3 m: Looks good.
- Relative humidity at ~3 m: Looks good. Minor adjustment made for drift in sensitivity starting in 2023.
- Apogee IRT surface temperature: Looks good.
- Snow temperature at 0.2 and 0.5 m heights: Look good. Average snow temperature calculated.
- Soil temperature at 0.005, 0.15 and 0.5 m depths: Look good.
- Snow depth: Looks good. Preliminary estimate of daily and hourly depths need filtering for signal noise due to thermal effects, blowing snow, rain etc.
- Rain gauge: Not working from 2021 onwards.
- Solar radiation - incident: Missing from daily output starting in June 2021 after replacement of sensor and program revision. Hourly data are available and used to calculated daily values. Correction for snow on sensor using reflected data. Algorithm added to hourly files to evaluate shading and sensitivity drift of sensor.
- Solar radiation - reflected: Missing from daily output starting in June 2021 after replacement of sensor and program revision. Hourly data are available and used to calculated daily values.
- Soil moisture: A TDR probe was added in June 2021. Data look good.
- Wind speed at ~3 m: Looks good. May be decreasing over time as surrounding forest canopy increases.
- Wind direction at~3 m: Looks good.
- Vapour pressure and vapour pressure deficit: Calculated from air temperature and relative humidity on hourly and daily basis.

PC

- Air temperature at ~3 m: Looks good.
- Relative humidity at ~ 3m: Looks good. No adjustment made for drift in sensitivity.
- Apogee IRT surface temperature: Looks good.
- Snow temperature at 0.2 and 0.5 m heights: 0.2 m looks good.0.5 m sensor is on the ground since 2019. Sensor is OK, needs a new stick to hold it at 0.5 m. Average snow temperature calculated.
- Soil temperature at 0.005, 0.15 and 0.5 m depths: Look good.
- Snow depth: Looks good. Preliminary estimate of daily and hourly depths need filtering for signal noise due to thermal effects, blowing snow, rain etc.
- Soil moisture: A TDR probe was added in June 2021. Data look good.
- Vapour pressure and vapour pressure deficit: Calculated from air temperature and relative humidity on hourly and daily basis.

PJ

- Air temperature at ~3 m: Looks good.
- Relative humidity at ~3 m: Looks good. No adjustment made for drift in sensitivity.
- Apogee IRT surface temperature: Intermittent failure since 2019.
- Snow temperature at 0.2 and 0.5 m heights: Look good. Average snow temperature calculated.
- Soil temperature at 0.005, 0.15 and 0.5 m depths: Look good.
- Snow depth: Looks good. Preliminary estimate of daily and hourly depths need filtering for signal noise due to thermal effects, blowing snow, rain etc.
- Rain gauge: Not working from 2023 onwards.
- Solar radiation - incident: Missing from daily output starting in September 2021 with replacement of sensor and revision to programming. Hourly data are available and used to calculated daily values. Correction for snow on sensor using reflected data. Algorithm added to hourly files to evaluate shading and sensitivity drift of sensor.
- Solar radiation - reflected: Looks good.
- Soil moisture: A TDR probe was added in June 2021. Data look good.
- Wind speed at ~3 m: Looks good. May be decreasing over time as surrounding forest canopy increases.
- Wind direction at~3 m: Looks good.
- Vapour pressure and vapour pressure deficit: Calculated from air temperature and relative humidity on hourly and daily basis.

PK

- Missing data February to July 2021 and May to October 2024 due to battery failure.
- Air temperature at ~3 m: Looks good.
- Relative humidity at ~3 m: Looks good. Minor adjustment made for drift in sensitivity.
- Apogee IRT surface temperature: Looks good.
- Snow temperature at 0.2 and 0.5 m heights: Look good. Average snow temperature calculated.
- Soil temperature at 0.005, 0.15 and 0.5 m depths: Look good.
- Snow depth: Looks good. Preliminary estimate of daily and hourly depths need filtering for signal noise due to thermal effects, blowing snow, rain etc.
- Rain gauge: Looks good.
- Solar radiation - incident: Missing from daily output starting in July 2021 when a new data logger and sensor were installed and program revised. Hourly data are available and used to calculated daily values. Correction for snow on sensor using reflected data. Data for 2021 to 2024t are too low (possibly a programming error). No data May 2024 onwards. Added algorithm to hourly files to evaluate shading and sensitivity drift of sensor.
- Solar radiation - reflected: Looks good until May 2024, then no data. Missing from daily output starting in July 2021 when a new data logger and sensor were installed and program revised. Hourly data are available and used to calculated daily values.
- Soil moisture (depth unknown): A TDR probe was added in July 2021. Data look good.

- Wind speed: Starts in October 2024. Looks good.
- Wind direction: Starts in October 2024. Only a daily average value recorded. No hourly data.
- Vapour pressure and vapour pressure deficit: Calculated from air temperature and relative humidity on hourly and daily basis.

PL

- Hourly and daily data evaluated and summarized in single spreadsheet for a year. Graphs of daily time series.
- The 2019 spreadsheet includes a graphical comparison of daily data with station PB.
- Air temperature at ~2 m: Looks good.
- Relative humidity at ~2 m: Looks good. No adjustment made for drift in sensitivity.
- Snow depth: Looks good. Preliminary estimate of daily and hourly depths need filtering for signal noise due to thermal effects, blowing snow, rain etc. May need to revise reference distance to ground by 0.05 to 0.1 m to get true absolute depth depending on year.
- Rain gauge: Looks good.
- Solar radiation - incident: Looks good but no correction for snow on sensor in winter. Algorithm added to hourly worksheet to evaluate shading and sensitivity drift of sensor. Missing data June 2020 to Aug 2022.
- Soil moisture at 0.1 and 0.2 m depths: Look good.
- Wind speed at ~2 m: Looks good. May be decreasing over time as surrounding forest canopy increases.
- Wind direction at~2 m: Looks good. Need to check orientation. May be 180 degrees out, or was initially installed out by 180 degrees.
- Vapour pressure and vapour pressure deficit: Calculated from air temperature and relative humidity on hourly and daily basis.

Geonor

- Hourly and daily data evaluated and summarized in single spreadsheet for a year. Graphs of hourly and daily time series of original and filtered data. Precipitation calculated from filtered data.
- Hourly and daily precipitation calculated from respective signals. Data filtered for gauge recharge, to remove noise and account for evaporation loss. Data look good. Signal noise increases over time. There are pulses that rise in value for an hour or two and then drop back to previous values. This would indicate precipitation at times when there is none. Adjustment for this could be improved.